VIDYAPEETH

BATCH CODE: 12-AN181MA 2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNITS AND MEASURMENT

Lecture No. 05

By – Navneet Shrivastav

Today's Goal

Subtopic Application of Dimensional Analysis

Subtopic Principle of Homogenity

Subtopic

REVISION OF LAST CLASS

Class Work :

1. Take out a sheet and write Dimensions of any 10 PQs
2. Write PQs having same dimensions

$$F = \frac{\Delta P}{\Delta t}$$

$$\Delta P = F \cdot \Delta t$$

Impulse :- $I = F \cdot dt$ (Ns)

$I = F \cdot \Delta t$

$$\boxed{I = \Delta P}$$

Dimensionless PQs :-

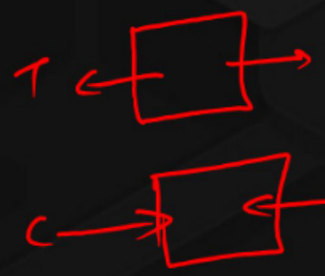
$\rho \rightarrow$ Density

1. Relative Density (Unitless) $= \dfrac{\rho_s}{\rho_w} = \dfrac{kg/m^3}{kg/m^3} = 1$ = Dimensionless.

2. Refractive Index (Unitless) $= \dfrac{C}{V} = \dfrac{m/s}{m/s} = 1$

3) Plane Angle (radian) $= \dfrac{\text{length of Arc (m)}}{\text{Radius (m)}} = 1$ (Dimless But Not Unitless)

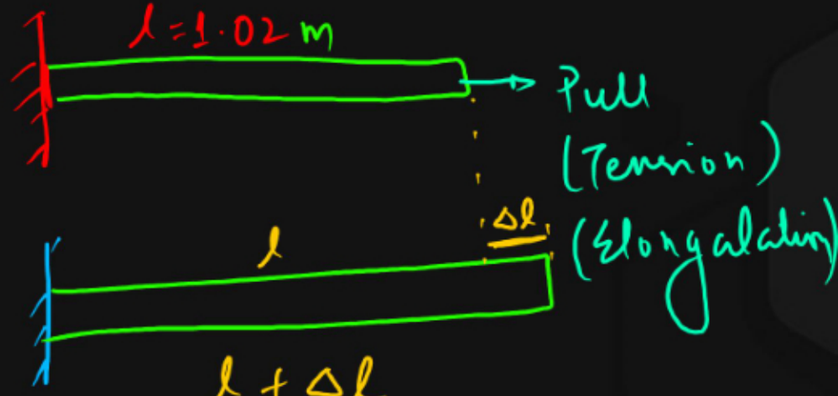
4) STRAIN $= \dfrac{\Delta l\,(m)}{l\,(m)} = 1$



Tensile Force → Tension → Elongation

compressive Force

always act Away from Body



Change in length

$$\text{Strain} = \frac{\text{change in length}}{\text{Original length}} = \frac{\Delta l\ (m)}{l\ (m)}$$

(Unitless as well as Dimensionless)

(5) All Trigonometric function
Logarithmic fun^c
Exponential fun^c

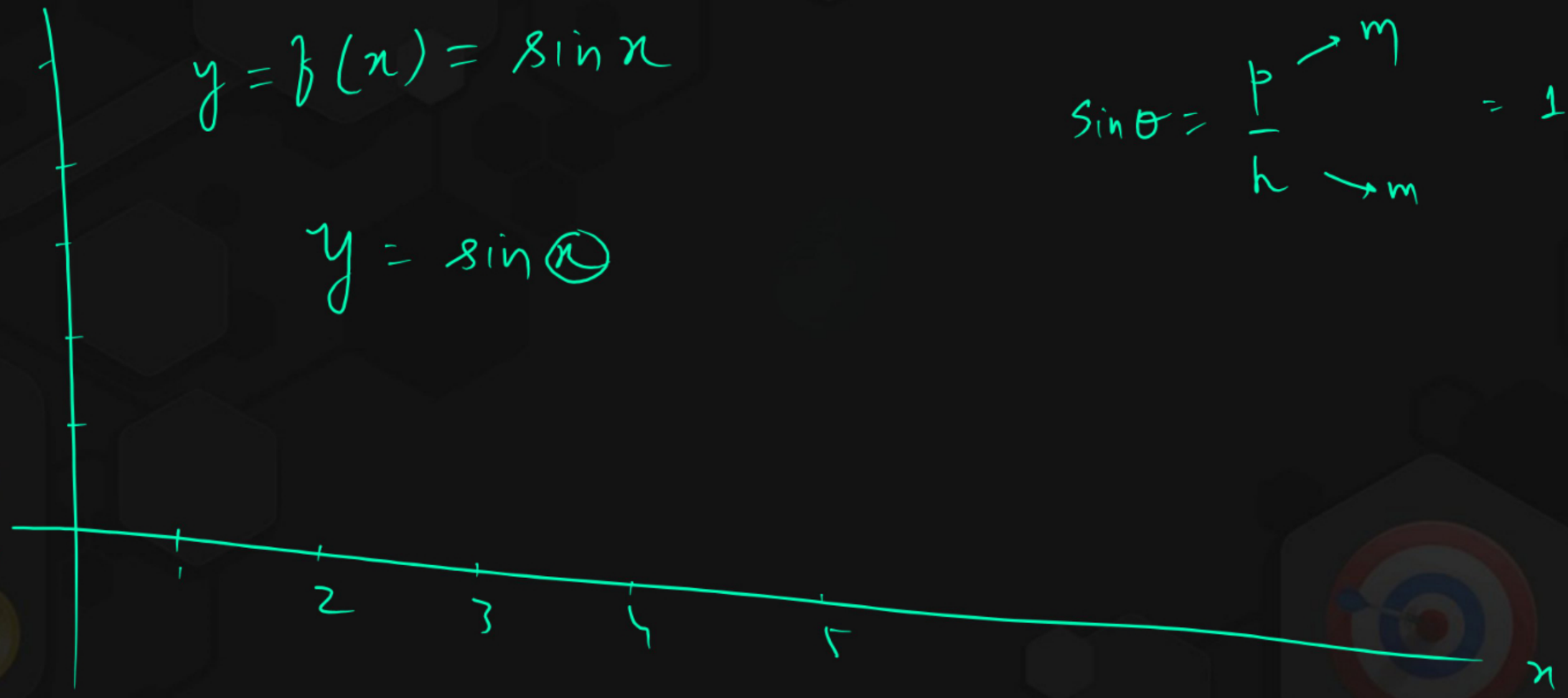
} Dimensionless.

$\log x$ | e^x

$y = f(x) = \sin x$

$y = \sin x$

$\sin\theta = \frac{p}{h} = 1$



APPLICATION OF DIMENSIONAL ANALYSIS

1. Principle of HOMOGENITY

2. To convert a physical quantity from one system of measurement to another system of measurement.

3. Deducing relationship between physical quantities.

Principle of HOMOGENITY

$PQ = n \cdot u$

$KE = \frac{1}{2} m v^2$

$ML^2T^{-2} = ML^2T^{-2}$

$s = ut + \frac{1}{2} at^2$

$[LHS] = [RHS]$

$[L]$

$ut + \frac{1}{2} at^2$

$LT^{-1} \cdot T + LT^{-2} T^2$

$[L] + [L]$

$\Downarrow$

$[L]$

Topic: Homework-1

- DPP

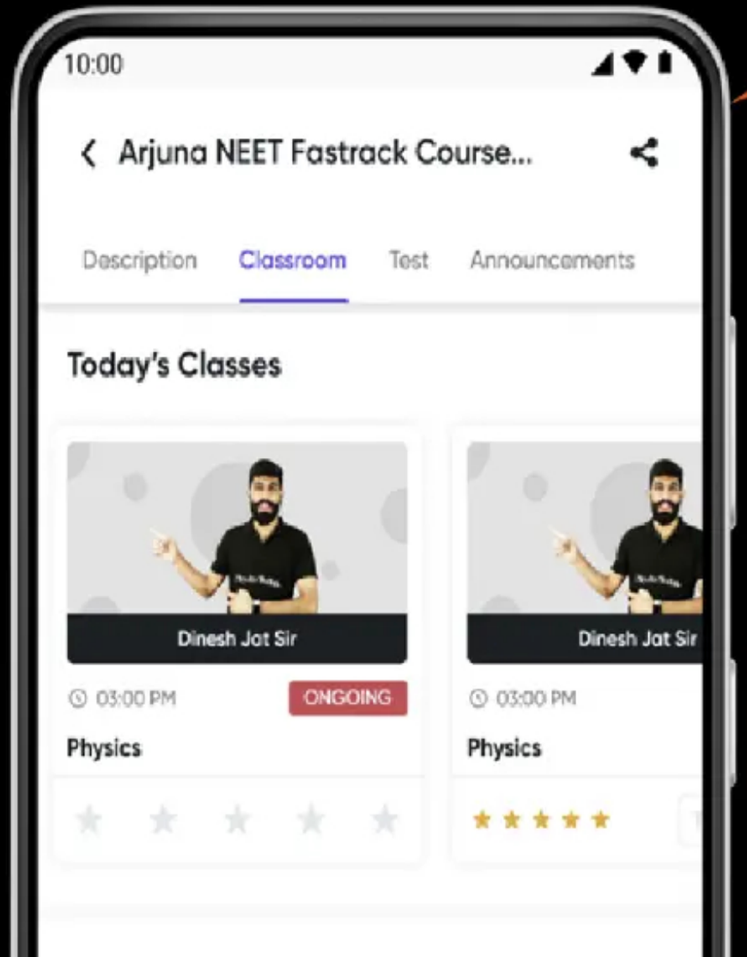
DPP NO	Question No
DPP -1	
DPP -2	

- Module
  - Exercise Name

Module Page No	Question No
Page No- 21	
Page No -22	

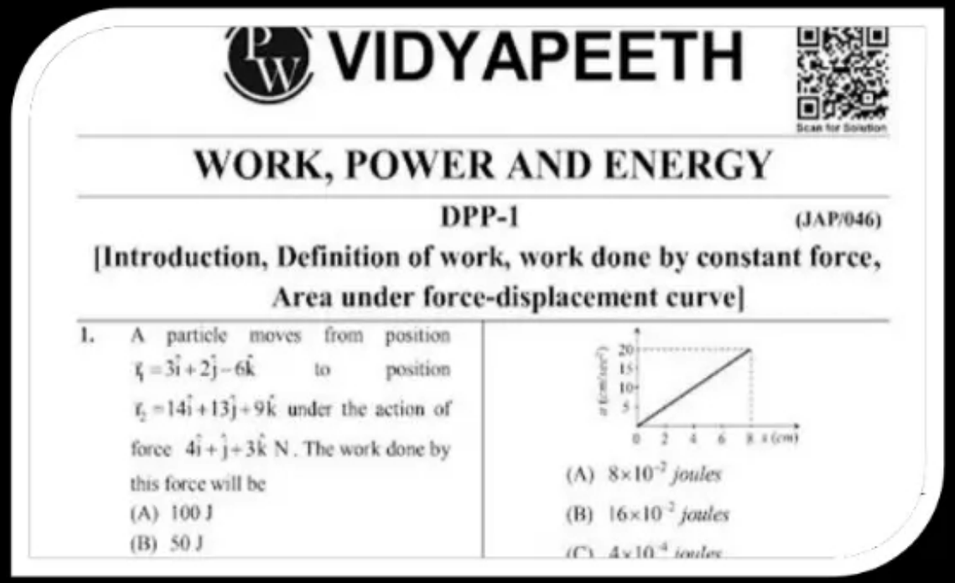
★★★★★

Rate your today's class in the Batch



"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



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Scan for Solution

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} *joules*

(B) 16×10^{-2} *joules*

Thank You!!!!

Check your performance through Student Dashboard